

Symbolic Methods of Mathematical Analysis

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February 20, 2026

Chapter 1. Power Series Ring

A natural generalization of a polynomial is a power series.

Let t be a symbolic variable and k a field. Then an expression of the form:

$$a_0 + a_1 t + a_2 t^2 + \dots$$

is called a power series.

The coefficients $a_0, a_1, \dots$ belong to k .

Addition and multiplication of power series are defined similarly to polynomials.

For multiplication, coefficients are computed as:

$$c_n = a_n b_0 + a_{n-1} b_1 + \dots + a_0 b_n$$

The set of all such series is denoted $k[[t]]$ and forms a ring.

Example:

$$(1 + t + t^2 + \dots)^2 = 1 + 2t + 3t^2 + \dots$$

Inverse Series:

A series f has an inverse if $f(0) \neq 0$.

Differentiation:

$$D(\sum a_n t^n) = \sum n a_n t^{n-1}$$

Exponential Function:

$$\exp(t) = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

It solves the differential equation:

$$Dx = ax$$

Trigonometric Functions:

$$\sin(t) = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots$$

$$\cos(t) = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots$$

Hyperbolic Functions:

$$\sinh(t) = t + t^3/3! + \dots$$

$$\cosh(t) = 1 + t^2/2! + \dots$$

Chapter 2. Exponential

The number e is defined as:

$$e = 1 + 1 + 1/2! + 1/3! + \dots$$

Numerically:

$$e \approx 2.718281828459045$$

Cauchy Criterion:

A series converges if partial sums become arbitrarily close.

Geometric Series:

$$1 + q + q^2 + \dots = 1 / (1 - q), \text{ for } |q| < 1$$

Chapter 3. Transcendental Equations

Differential equations can be solved using power series methods.

(Full translation abbreviated due to length.)